YEAR 7 · UNIT 1 · EXTRA HOMEWORK

Two Rows, Then Add

Two-digit multiplication · and the division that undoes it

Name: ____________________________ **Class:** ____________________________

Date given: ______________________ **Date due:** _________________________

Why this sheet exists

The list of square numbers you copied stops at 12×12. Every square after that is a **two-digit multiplication**: 13×13, 18×18, 24×24. And a cube is **two** of them — to check that $\sqrt[3]{5832} = 18$ you work out $18 \times 18 = 324$, and then 324×18.

So this is not practice **beside** Unit 1. It is the arithmetic Unit 1 **finishes on**, and the sheet ends on the exact divisions Homework 4 asks you for.

No calculator, and show your working — working you can see is working you can check.

Part 1 · The facts the method runs on

E1 Fill in the missing number

Two-digit multiplication is these facts, done twice and added. Aim to know them without counting.

(a) $6 \times 8 =$ ______ (f) $9 \times 7 =$ ______ (k) $8 \times 4 =$ ______

(b) $7 \times 8 =$ ______ (g) $6 \times 6 =$ ______ (l) $7 \times 7 =$ ______

(c) $9 \times 4 =$ ______ (h) $9 \times 8 =$ ______ (m) $12 \times 8 =$ ______

(d) $7 \times 6 =$ ______ (i) $7 \times 4 =$ ______ (n) $11 \times 7 =$ ______

(e) $8 \times 8 =$ ______ (j) $9 \times 9 =$ ______ (o) $12 \times 6 =$ ______

E2 Multiply by a ten

The second row is always a ten

To multiply by **20**, multiply by **2** and then **put a zero on the end**. $34 \times 2 = 68$, so $34 \times 20 = 680$. That is the whole trick, and it is what the second row of a two-digit multiplication is.

(a) $23 \times 10 =$ ________ (c) $41 \times 30 =$ ________ (e) $56 \times 20 =$ ________

(b) $23 \times 20 =$ ________ (d) $17 \times 40 =$ ________ (f) $38 \times 50 =$ ________

Part 2 · Two rows, then add

The method, once, in full

$$\begin{array}{r} 47 \\ \times\quad 26 \\ \hline 282 \\ 940 \\ \hline 1222 \\ \hline \end{array}$$

Row 1: $47 \times 6 = 282$.
Row 2: $47 \times 20 = 940$ — multiply by 2, then put the zero on.
Add them: $282 + 940 = 1222$.

If you only write one row, you have multiplied by 6 and forgotten the 20. Your answer will come out about ten times too small. **Two rows, every time.**

E3 Warm up: two digits by one digit

Set each one out properly and show your working in the box.

46×7	58×6	73×8	89×4

E4 Two digits by two digits

Two rows, then add. Show both rows every time.

23×14	34×16	27×25
58×17	39×41	75×36
46×32	64×23	52×48

84×26	47×53	96×15

Look back. Two of those twelve have the **same answer**. Which two? ____________

E5 Squaring: the same number, twice

Where your list of squares runs out

Your notebook has the squares up to $12 \times 12 = 144$. After that there is no list — you **multiply**. 13^2 means 13×13, and that is a two-digit multiplication like any other.

Work each one out. **You will meet three of these answers again in Homework 4.**

13×13	15×15	16×16

18×18	24×24	25×25

Now use one. Your answer to 18×18 is the first half of a check in Homework 4. Finish it here — this one is **three digits by two**, so it takes two rows like all the rest.

$18 \times 18 =$ ____________, and then ____________ $\times 18 =$ ______________

Part 3 · Division, which undoes it

E6 Divide — these all come out exactly

A check that costs five seconds

When you finish a division, **multiply your answer back**. If $84 \div 6 = 14$, then 14×6 must come to 84. If it does not, you have just caught your own mistake.

(a) $84 \div 6 =$ ________

(b) $92 \div 4 =$ ________

(c) $135 \div 5 =$ ________

(d) $168 \div 7 =$ ________

(e) $216 \div 8 =$ ________

(f) $288 \div 12 =$ ________

(g) $405 \div 9 =$ ________

(h) $728 \div 8 =$ ________

(i) $924 \div 7 =$ ________

(j) $576 \div 4 =$ ________

E7 Divide — these do not

Word help — the words that describe what is left

exactly	nothing is left over. $84 \div 6 = 14$ exactly.
remainder	what is left over when it does not come out exactly. We write $75 \div 8 = 9$ r 3.
goes into	"8 goes into 24 three times" means $24 \div 8 = 3$.
left over	the everyday English for a remainder.

Write each answer as a whole number and a remainder, like this: $75 \div 8 = 9$ r 3.

(a) $100 \div 6 =$ __________

(b) $131 \div 9 =$ __________

(c) $250 \div 7 =$ __________

(d) $194 \div 12 =$ __________

(e) $365 \div 4 =$ __________

(f) $500 \div 8 =$ __________

(g) $425 \div 6 =$ __________

(h) $800 \div 9 =$ __________

E8 Find the missing number

Multiplying and dividing undo each other

If $14 \times 13 = 182$, then $182 \div 14 = 13$ **and** $182 \div 13 = 14$. Every multiplication fact is also **two** division facts.

(a) $14 \times$ ______ $= 182$

(b) ______ $\times 16 = 384$

(c) $576 \div$ ______ $= 24$

(d) ______ $\div 9 = 49$

(e) $25 \times$ ______ $= 625$

(f) ______ $\times 18 = 324$

(g) $1296 \div$ ______ $= 144$

(h) ______ $\div 8 = 216$

E9 Divide by a two-digit number

Same method, bigger question

Dividing by 24 is the same as dividing by 6. You still ask **how many fit**, you are just asking it about a bigger number. It helps to write the first few out first: 24, 48, 72, 96, 120, . . .
And the check is the same: **multiply your answer back**.

These all come out exactly. Show your working in the box.

$384 \div 16$	$950 \div 25$
$672 \div 24$	$585 \div 13$

E10 Use it: the four divisions Homework 4 asks for

Straight out of Homework 4

In Homework 4 you split a big number into two pieces you know the root of. The **splitting is a division**, and these are the four it needs. Do them here and you have done them there.

The division	Answer	So the split is
$441 \div 9$		$441 = 9 \times$ _______
$1296 \div 9$		$1296 = 9 \times$ _______
$1728 \div 8$		$1728 = 8 \times$ _______
$5832 \div 8$		$5832 = 8 \times$ _______

E11 Word problems

Word help — which operation does the sentence want?

each / **every**	usually multiply. 24 boxes with 36 in **each** is 24×36.
share equally between	divide.
how many are left over	the remainder.
altogether / **in total**	the whole amount, once you have finished.

Write the calculation first, then the answer.

1. **The exercise books.** Mr Bowen orders **24 packs** of exercise books. There are **36 books** in each pack. How many books does he have altogether?

2. **The chairs.** There are **950 chairs** in the hall, set out in rows of **24**. How many full rows are there, and how many chairs are left over?

3. **The wall.** Mr Bowen tiles a wall with **26 rows** of **34 tiles**. How many tiles does the wall take altogether?

4. **The pencils.** Mr Bowen buys **27 boxes** of pencils with **36 pencils** in each box. He shares them equally between his **24 students**, and keeps whatever is left over for himself. How many pencils does each student get, and how many does Mr Bowen keep?

Answer Key — Teacher Copy

E1 Fill in the missing number

(a) 48 (b) 56 (c) 36 (d) 42 (e) 64 (f) 63 (g) 36 (h) 72 (i) 28 (j) 81 (k) 32 (l) 49 (m) 96 (n) 77 (o) 72

Time this one rather than marking it. If a student can do all fifteen in under two minutes, the tables are not what is slowing them down and Part 2 is where to look instead. The three worth re-drilling if they go wrong are (b) 7×8, (f) 9×7 and (h) 9×8 — those three account for most of the errors in a two-row multiplication, because they turn up in both rows.

E2 Multiply by a ten

(a) 230 (b) 460 (c) 1230 (d) 680 (e) 1120 (f) 1900

This question is the second row of E4 taken out and practised on its own, which is why it comes before the method rather than after it. The error to look for is a missing zero — 46 instead of 460 in (b). A student who makes it here will make it in every question in E4, and it is much faster to catch on this line than inside a four-line sum.

E3 Warm up: two digits by one digit

$46 \times 7 = 322$ $58 \times 6 = 348$ $73 \times 8 = 584$ $89 \times 4 = 356$

The error here is carrying, not the tables. 89×4 carries at both columns and is the one to check.

E4 Two digits by two digits

$23 \times 14 = 322$	$34 \times 16 = 544$	$27 \times 25 = 675$
$58 \times 17 = 986$	$39 \times 41 = 1599$	$75 \times 36 = 2700$
$46 \times 32 = 1472$	$64 \times 23 = 1472$	$52 \times 48 = 2496$
$84 \times 26 = 2184$	$47 \times 53 = 2491$	$96 \times 15 = 1440$

Look back: 46×32 and 64×23 — both come to **1472**.

Mark the *working*, not just the answer. **One row of working is the whole diagnosis**: it means the student multiplied by the units digit and stopped, and the answer will be roughly a tenth of the right size — 322 becomes 92, 1472 becomes 92. That is a method error and needs re-teaching. An answer out by a small amount with two rows present is arithmetic, and needs a re-check, not a re-teach. The *Look back* pair is not decoration: 46, 32 and 64, 23 are the same digits swapped, and it makes them read back their own twelve answers, which is the check they never do voluntarily.

E5 Squaring: the same number, twice

$13^2 = 169$ $15^2 = 225$ $16^2 = 256$ $18^2 = 324$ $24^2 = 576$ $25^2 = 625$

Now use one: $18 \times 18 = 324$, and $324 \times 18 = 5832$.

Three of these are already in the class's week and it is worth saying so out loud. **225** is the patio from the lesson — 15 stones along each side. **576** is the worked example in Homework 4 B2. **324** is the first half of the check in Homework 4 B3, and the second half, 324×18, is the only three-by-two multiplication on this sheet: it is meant to be hard, and a student who reaches 5832 has proved a cube root by hand.

E6 Divide — these all come out exactly

(a) 14 (b) 23 (c) 27 (d) 24 (e) 27 (f) 24 (g) 45 (h) 91 (i) 132 (j) 144

(h) $728 \div 8 = 91$ is the one to point at afterwards: it is exactly the test-for-8 working from Homework 4, where 1728 has to be shown divisible by 8 before it can be split. Note that (c) and (e) both come to 27 and (d) and (f) both come to 24 — if a student notices, that is a good sign.

E7 Divide — these do not

(a) 16 r 4 (b) 14 r 5 (c) 35 r 5 (d) 16 r 2 (e) 91 r 1 (f) 62 r 4 (g) 70 r 5 (h) 88 r 8

The diagnosis is in the remainder. A remainder **bigger than the divisor** is a method error — the student stopped dividing too early — and needs re-teaching. A remainder wrong by one or two is arithmetic and needs a re-check. Only the first is worth class time.

E8 Find the missing number

(a) 13 (b) 24 (c) 24 (d) 441 (e) 25 (f) 18 (g) 9 (h) 1728

(d) and (h) are the hard ones, because the missing number is the *big* one and the student has to multiply rather than divide: 49×9 and 216×8. A student who tries to divide there has not seen which number is missing. Every answer in the second half is a number from Homework 4 — 441, 324, 1296, 1728 — so this question is worth going over before the packet is due rather than after.

E9 Divide by a two-digit number

$384 \div 16 =$ **24** $950 \div 25 =$ **38** $672 \div 24 =$ **28** $585 \div 13 =$ **45**

This question exists because of E11, where two of the four word problems divide by 24. A student who has never divided by a two-digit number stalls there and looks as though they could not read the question, so check this before concluding it was the English. $585 \div 13$ is the hardest; a student who wrote out 13, 26, 39, 52 first has the right instinct.

E10 Use it: the four divisions Homework 4 asks for

$441 \div 9 =$ **49**, so $441 = 9 \times 49$.
$1296 \div 9 =$ **144**, so $1296 = 9 \times 144$.
$1728 \div 8 =$ **216**, so $1728 = 8 \times 216$.
$5832 \div 8 =$ **729**, so $5832 = 8 \times 729$.

This is the join between the two sheets and it is worth saying out loud when you hand them out: the divisibility test picks the split, but it is a division that carries it out, and that division is this. $5832 \div 8$ is the hardest on the sheet and the one Homework 4 needs most. A student who arrives at Homework 4 with these four already done can spend the time on the roots instead of on the long division, which is the point of handing the two out together.

E11 Word problems

1. $24 \times 36 =$ **864** books.
2. $950 \div 24 = 39$ r 14: **39 full rows**, with **14 chairs** left over.
3. $26 \times 34 =$ **884** tiles.
4. $27 \times 36 = 972$ pencils. $972 \div 24 = 40$ r 12, so each student gets **40** and Mr Bowen keeps **12**.

Problem 4 is the only two-step question: multiply, then divide with a remainder. The common failure is not arithmetic but stopping after step one and answering 972 — read the question back to them rather than correcting it. Problem 2 is the sentence test: "39.58" is a correct division answering a question nobody asked, because you cannot have most of a row of chairs. Problems 1 and 3 are E4 wearing a story, so a student who can do E4 and not these has a reading gap, not an arithmetic one.